

ECO387C - Fall 2024

Problem Set 3

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You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

Question 1 - Cake-eating problem

Consider an agent who is endowed with a certain amount of cake g_0 at time 0. The agent can consume a portion of the cake and save the rest for future consumption. In other words, the agent seeks to maximize

$$\max_{c_t} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad \text{where} \quad u(c_t) = \frac{c_t^{1-\gamma}}{1-\gamma} \quad \text{and} \quad 0 < \beta < 1$$

subject to

$$g_{t+1} = g_t - c_t \quad g_0 > 0 \text{ is given} \quad \text{and} \quad 0 \leq c_t \leq g_t$$

- a Set up the Lagrangian.
- b Derive the agent's Euler equation.
- c Set up the Bellman equation.
- d Derive the agent's Euler equation from the Bellman equation.
- e Solve for the agent's optimal policy functions: consumption (c_t^*) and saving (g_{t+1}^*).
- f Show that the value function takes the following form:

$$V(g_t) = u(g_t)(1 - \beta^{1/\gamma})^{-\gamma}$$

Question 2 - Log-linearization

- a Prove that log-linearizing the left-hand side of each equation yields the corresponding right-hand side. $\widehat{X}_t$ indicates the percentage deviation from the trend ($\widehat{X}_t = \frac{X_t - \bar{X}}{\bar{X}}$ where $\bar{X} \neq 0$)

a) $\widehat{aX_t} = \widehat{X_t}$

b) $\widehat{X_t^a} = a\widehat{X_t}$

c) $\widehat{X_t Y_t} = \widehat{X_t} + \widehat{Y_t}$

d) $\frac{\widehat{X_t}}{\widehat{Y_t}} = \widehat{X_t} - \widehat{Y_t}$

e) $\widehat{\frac{X_t}{X_t + Y_t}} = \frac{\bar{X}}{\bar{X} + \bar{Y}} \widehat{X_t} + \frac{\bar{Y}}{\bar{X} + \bar{Y}} \widehat{Y_t}$

f) $\widehat{\frac{X_t}{X_t - Y_t}} = \frac{\bar{X}}{\bar{X} - \bar{Y}} \widehat{X_t} - \frac{\bar{Y}}{\bar{X} - \bar{Y}} \widehat{Y_t}$

- b Based on the results above, log-linearize the following equations. Assume a non-zero, constant trend for each variable ($\bar{X}_t = \bar{X}$). Greek letters without subscript t are constant parameters.

a) $Y_t = C_t + I_t + G_t$

b) $K_{t+1} = I_t + (1 - \delta)K_t$

c) $Q_t = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma}$

d) $\frac{W_t}{P_t} = \alpha \frac{A_t}{N_t^{1-\alpha}}$